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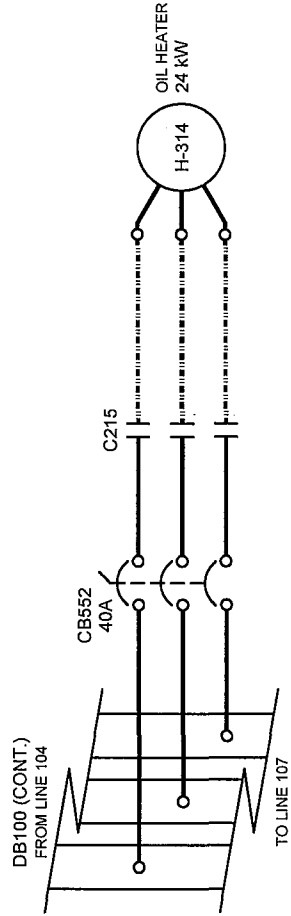
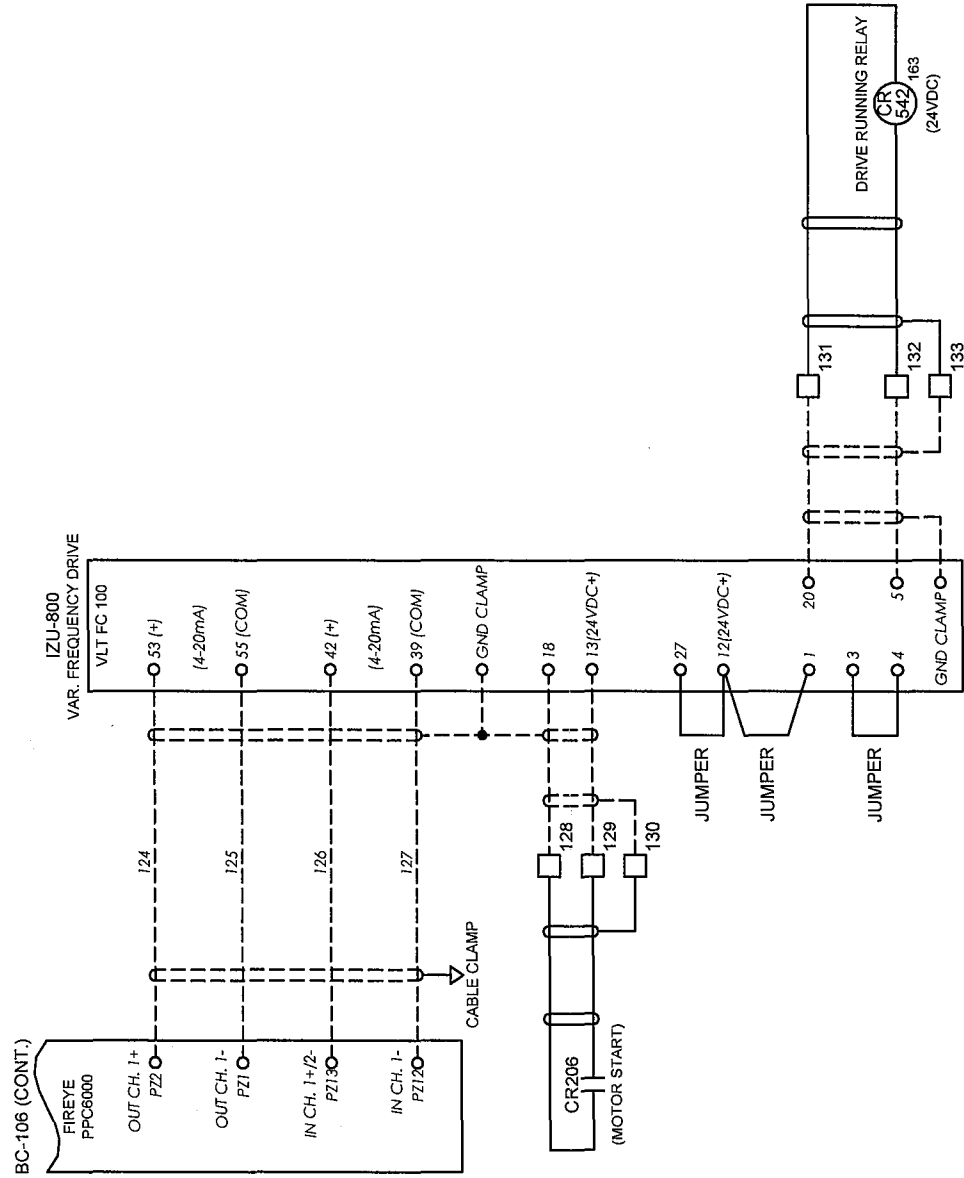
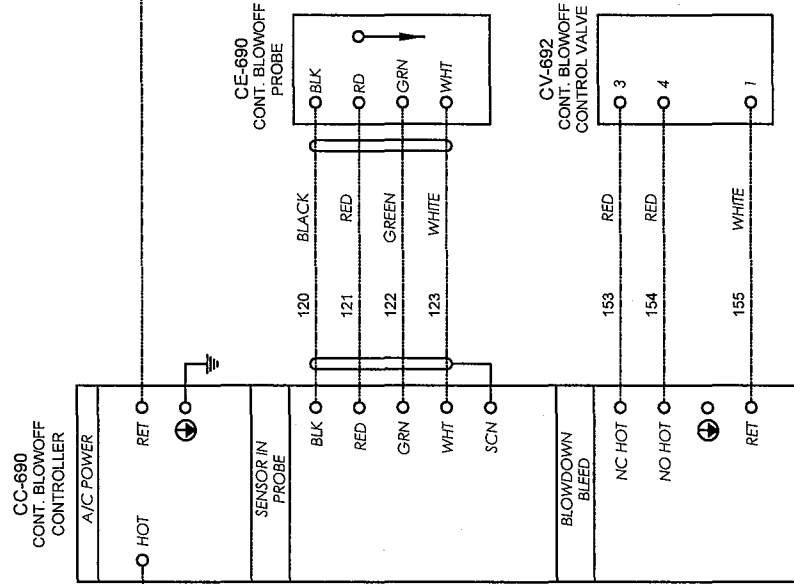
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SCALE NONE	JOHNSTON BOILER COMPANY FERRYSBURG, MICHIGAN 49409	SO#
DATE 9-19-17		
DRAWN T. NINK	WIRING SCHEMATIC YB110/YZ300/PPC6000 - HG	REV
CHECKED		
APPROVED T. JN	DWG. NO. 1137730801	(PAGE 5 OF 5)

Johnston Boiler Company
Bill of Material

Assembly P/N: 11377X1

Description: Ship Loose Items

Dwg. No: -

Tag No.	Qty.	Service	Rev.	JBC	Name	Specification	Comment
PI-208	1	Regulated Gas Pressure Gauge		011279	GAGE PRESSURE 0-10# 2.50	ASHCROFT #25-1490A-02L-10 PSI 1/4	
PI-226	1	Burner Gas Pressure Gauge		011279	GAGE PRESSURE 0-10# 2.50	ASHCROFT #25-1490A-02L-10 PSI 1/4	
PI-514	1	Pilot Pressure Gauge		011276	GAGE PRESSURE 0-15" WC 2.50	ASHCROFT 25-1490A-02L-15-IW	
PI-602	1	Steam Pressure Gauge		013043	GAGE PRESSURE 0-300# 8.50	ASHCROFT 85-1010A-02L-300PSI 1/4" CONN	
TI-644	1	Stack Thermometer		012536	THERMOMETER 5.0 100-800 DGF	REOTEMP SS-060-1-F79 W/6" STEM	
PSV-650	3	Safety Valves		024403	VALVE SR 3.00X4.00 24008	KUNKLE 6252FLK01-AS0150	SP = 150 psig. CAP = 24,008 lbs/hr ea.
-	3	Flange Kits (Safety Valves)		034441	FLANGE KIT STD 300#RF 3.00	INCLUDES STUDS NUTS & GASKET - FLEX	
-	1	Feedwater Control Valve		034732	VALVE FEEDWATER ASSY 1.50" NPT	JOHNSTON JBFV-150-1HXS	
-	1	Slow Opening Blowdown Valve		012658	VALVE BLOWDOWN 2.000 NPT SO	EVERLASTING 4060-A	
IZU-800	0	Variable Frequency Drive (Blower Motor)		035974	INVERTER PKG NEMA12 W/DISCONN	DANFOSS/FIREYE NXVFD1004 100HP N	Note: Quantity of (1) supplied.
-	1	Flue Brush		010274	BRUSH FLUE 2.50 CARBON STEEL	TANIS INC. #09826 - DOUBLE SPIRAL/STEM	
Notes							
Rev Data	No.	Date	By	Chk.	Description	Data Entry	
	0	20-Sep-17	TJN	TJN	-		

FIRST LETTERS – INITIATING OR MEASURED VARIABLE

A – ANALYSIS	J – POWER	T – TEMPERATURE
B – BURNER, COMUSTION	K – TIME	TD– DIFF. TEMPERATURE
C – CONDUCTIVITY	L – LEVEL	U – MULTIVARIABLE
D – DENSITY	M – NOT USED	V – VIBRATION
E – VOLTAGE	N – NOT USED	– MACHINERY ANALYSIS
F – FLOW RATE	O – NOT USED	W – WEIGHT, FORCE
FF– FLOW RATIO	P – PRESSURE/VACUUM	WD– DIFF. WEIGHT, FORCE
FQ– FLOW QUANTITY	PD– DIFF. PRESSURE	X – UNCLASSIFIED
G – NOT USED	Q – QUANTITY/NUMBER	Y – EVENT, STATE, PRESENCE
H – HAND	R – RESTRICTION	Z – POSITION, DIMENSION
I – CURRENT	S – SPEED/FREQUENCY	ZD– GAUGING, DEVIATION

TAG NUMBERS

PSH 228	– TYPICAL INSTRUMENT TAG NUMBER
228	– MAPICS BOM SEQUENCE NUMBER
PSH	– FUNCTIONAL IDENTIFICATION

SUCCEEDING LETTERS – INITIATING OR MEASURED VARIABLE

*A — * ALARM	*Q — * TOTALIZE/INTEGRATE
*AH — * ALARM HIGH	*R — * RECORD/PRINT
*AL — * ALARM LOW	*RC — * RECORDING CONTROLLER
*C — * CONTROLLER (BLIND)	*S — * SWITCH
*CV — * CONTROL VALVE (SELF–ACTUATED)	*SH — * SWITCH HIGH
*E — * ELEMENT	*SL — * SWITCH LOW
*G — * GLASS	*SV — * SAFETY OR RELIEF VALVE
*I — * INDICATOR	*T — * TRANSMITTER (BLIND)
*IC — * INDICATING CONTROLLER	*V — * VALVE
*L — * LIGHT	*W — * WELL
*O — * ORIFICE/RESTRICTION	*Y — * RELAY, COMPUTE, CONVERT
*P — * POINT/TEST CONNECTION	*Z — * DRIVER, ACTUATOR

(* REPRESENTS FIRST LETTERS FROM ABOVE LIST)

INSTRUMENT TYPE & LOCATION

	FIELD OR BOILER MOUNTED	PRIMARY CONTROL PANEL (NORMALLY ACCESSIBLE TO OPERATOR)	PRIMARY CONTROL PANEL (NORMALLY INACCESSIBLE TO OPERATOR)	AUXILIARY CONTROL PANEL (NORMALLY ACCESSIBLE TO OPERATOR)	AUXILIARY CONTROL PANEL (NORMALLY INACCESSIBLE TO OPERATOR)
DISCRETE INSTRUMENTS					
SHARED DISPLAY, SHARED CONTROL					
COMPUTER FUNCTION					
PROGRAMMABLE LOGIC CONTROL					

INSTRUMENT LINE SYMBOLS

INSTRUMENT SUPPLY OR CONNECTION TO PROCESS	
UNDEFINED SIGNAL	
PNEUMATIC SIGNAL	
ELECTRIC SIGNAL	
HYDRAULIC SIGNAL	
CAPILLARY TUBE	
ELECTROMAGNETIC OR SONIC SIGNAL (GUIDED)	
ELECTROMAGNETIC OR SONIC SIGNAL (UNGUIDED)	
INTERNAL SYSTEM LINK (SOFTWARE OR DATA LINK)	
MECHANICAL LINK	

DWG NO.
354B0013

REVISIONS
A 4-10-96

TE

ADD
ABBREVIATIONS.
CHG. CHECK VALVE
ABBREVIATION.



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THIS DRAWING ARE MADE WITH THE
UNDERSTANDING THAT THEY ARE
CONFIDENTIAL AND WILL NOT BE
USED IN ANY WAY DETRIMENTAL TO
THIS COMPANY'S INTEREST.

SCALE
NONE
DATE
3-22-96
DRAWN
TCE
CHECKED
APPROVED

JOHNSTON BOILER COMPANY
FERRYSBURG, MICHIGAN 49409

LEGEND SHEET

DWG. NO. 354B0013

(PAGE 1 OF 2)

REV A

VALVE SYMBOLS

GENERAL OR GATE	
ANGLE	
3-WAY	
4-WAY	
GLOBE	
BALL OR PLUG	
CHECK	
BUTTERFLY	
DIAPHRAGM	
ABREVIATION MAY BE USED FOR OTHER VALVE TYPES I.E. QUICK OPENING	



<u>ACTUATOR SYMBOLS</u>	
DIAPHRAGM, SPRING OPPOSED OR UNSPECIFIED	
DIAPHRAGM, PRESSURE BALANCED	
ROTARY MOTORIZED	
ELECTROHYDRAULIC	
SOLENOID	
HAND	
CYLINDER	
SAFETY OR RELIEF	



S

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ATM	—	ATMOSPHERE
BD	—	BLOWDOWN
CA	—	CORROSION ALLOWANCE
CHK	—	CHECK VALVE
CO	—	CHAIN OPERATED
CW	—	COOLING WATER
(FBO)	—	FURNISHED BY OTHERS
FC	—	FAIL CLOSED
FI	—	FAIL INDETERMINATE
FL	—	FAIL LOCKED (LAST POSITION)
FNPT	—	FEMALE NATION PIPE THREAD
FO	—	FAIL OPEN
FW	—	FEEDWATER
HOA	—	HAND—OFF—AUTO
IAS	—	INSTRUMENT AIR SUPPLY
ID	—	INSIDE DIAMETER
ILL	—	ILLUMINATED
Ic	—	INSULATION (COLD)
Ih	—	INSULATION (HOT)
Is	—	INSULATION (SAFETY)
kW	—	KILOWATT
MNPT	—	MALE NATIONAL PIPE THREAD
NC	—	NORMALLY CLOSED
NO	—	NORMALLY OPEN
NWL	—	NORMAL WATER LEVEL
RF	—	RAISED FACE FLANGE
SC	—	SAMPLE CONNECTION
SP	—	SET POINT
SG	—	SPECIFIC GRAVITY
TSO	—	TIGHT SHUTOFF

MISC. SYMBOLS		
INSTRUMENT WITH LONG TAG NUMBER		UNDEFINED INTERLOCK LOGIC
INSTRUMENTS SHARING COMMON HOUSING		TRAP
PILOT LIGHT		DIAPHRAGM SEAL
PURGE OR FLUSHING DEVICE		CAPPED CONNECTION
MANUAL RESET		PLUGGED CONNECTION
	CONNECTION OR LINE IDENTIFICATION	

PI
10-504

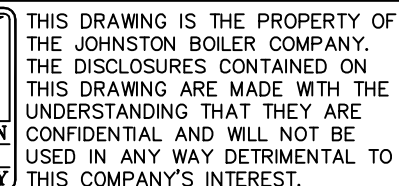


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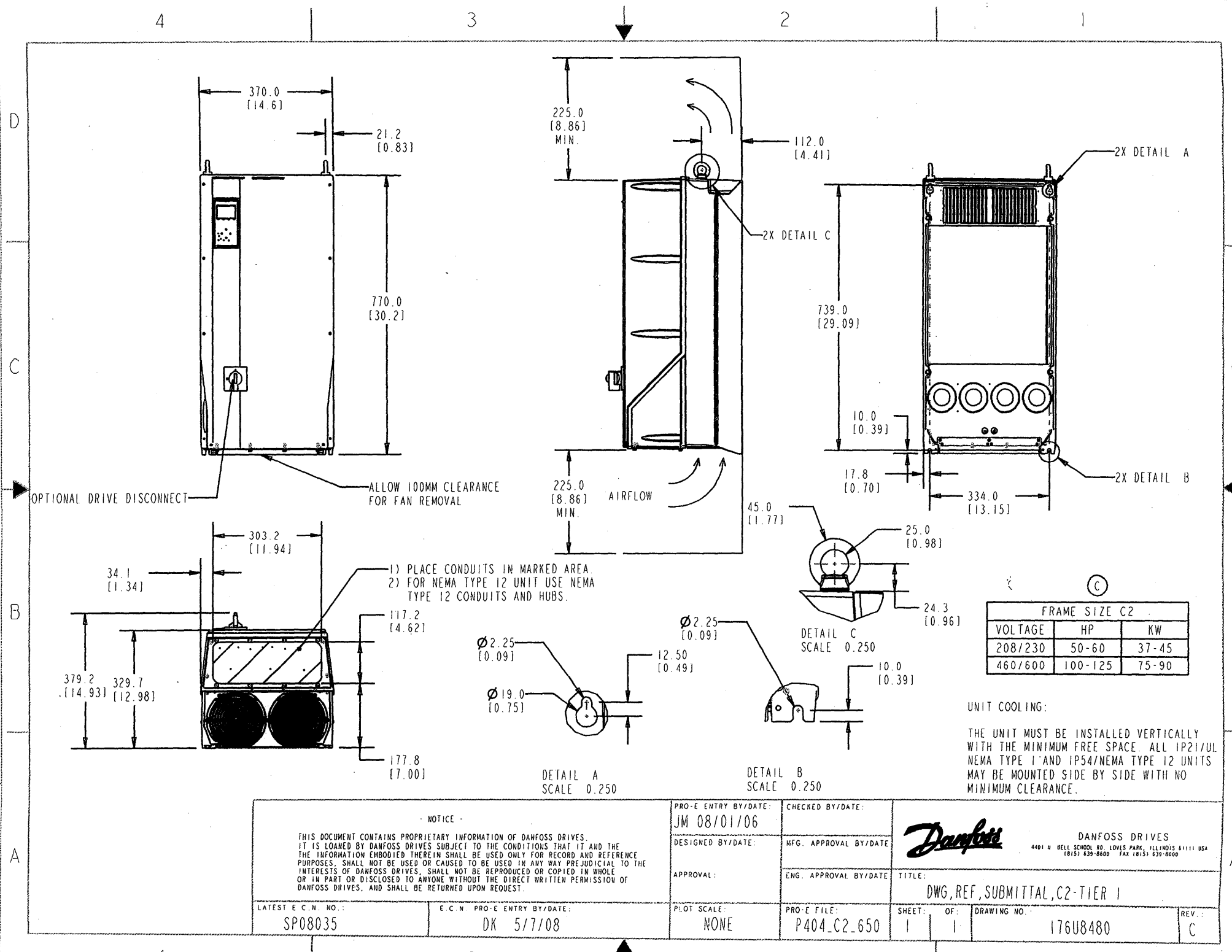


NOTE: FOR MORE DETAIL EXPLANATION OF SYMBOLS & THEIR USE OR FOR ADDITIONAL SYMBOLS SEE ISA STANDARD S5.1. WHERE DIFFERENCES OCCUR BETWEEN THIS LEGEND AND ISA S5.1, THIS LEGEND SHALL BE USED.



6	JOHNSTON BOILER COMPANY FERRYSBURG, MICHIGAN 49409
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LEGEND SHEET	
DWG. NO. 354B0013	(PAGE 2 OF 2) REV A



NOTICE THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF DANFOSS DRIVES. IT IS LOANED BY DANFOSS DRIVES SUBJECT TO THE CONDITIONS THAT IT AND THE INFORMATION EMBODIED THEREIN SHALL BE USED ONLY FOR RECORD AND REFERENCE PURPOSES. SHALL NOT BE USED OR CAUSED TO BE USED IN ANY WAY PREJUDICIAL TO THE INTERESTS OF DANFOSS DRIVES. SHALL NOT BE REPRODUCED OR COPIED IN WHOLE OR IN PART OR DISCLOSED TO ANYONE WITHOUT THE DIRECT WRITTEN PERMISSION OF DANFOSS DRIVES, AND SHALL BE RETURNED UPON REQUEST.		PRO-E ENTRY BY/DATE: JM 08/01/06		CHECKED BY/DATE:	
DESIGNED BY/DATE:		MFG. APPROVAL BY/DATE:		Danfoss DANFOSS DRIVES 4401 N. BELL SCHOOL RD. LOVELL PARK, ILLINOIS 61111 USA (615) 639-8600 FAX (615) 639-8000	
APPROVAL:		ENG. APPROVAL BY/DATE:		TITLE: DWG. REF. SUBMITTAL, C2-TIER 1	
LATEST E.C.N. NO.: SP08035	E.C.N. PRO-E ENTRY BY/DATE: DK 5/7/08	PLOT SCALE: NONE	PRO-E FILE: P404_C2.650	SHEET: OF: 1 1	DRAWING NO.: 176U8480
				REV.: C	



The
New York Blower
Company®

7660 QUINCY STREET—WILLOWBROOK, ILLINOIS 60527-5530
TEL: (630) 794-5700 • FAX: (630) 794-5778 • WEB: <http://www.nyb.com> • E-MAIL: nyb@nyb.com

INSTALLATION MAINTENANCE, OPERATING INSTRUCTIONS

IM-100

CENTRIFUGAL FANS AcF/PLR, AF, RTS



WARNING

THIS FAN HAS MOVING PARTS THAT CAN CAUSE SERIOUS BODILY INJURY. BEFORE OPERATING OR STARTING MAINTENANCE READ THE INSTALLATION AND MAINTENANCE INSTRUCTIONS AND THE AMCA SAFETY PRACTICES MANUAL PROVIDED WITH THIS FAN.

DURING OPERATION

1. KEEP BODY, HANDS, AND FOREIGN OBJECTS AWAY FROM THE INLET, THE OUTLET, AND THE OTHER MOVING PARTS OF THE FAN SUCH AS SHAFTS, BELTS, AND PULLEYS.
2. DO NOT OPERATE AT EXCESSIVE SPEEDS OR TEMPERATURES.

BEFORE STARTING MAINTENANCE WORK:
LOCK POWER SUPPLY IN OFF POSITION AND IMMOBILIZE FAN WHEEL.

98-0250

A WORD ABOUT SAFETY

The above **WARNING** decal appears on all **nyb** fans. Air moving equipment involves electrical wiring, moving parts, and air velocity or pressure which can create safety hazards if the equipment is not properly installed, operated and maintained. To minimize this danger, follow these instructions as well as the additional instructions and warnings on the equipment itself.

All installers, operators and maintenance personnel should study AMCA Publication 410, "Recommended Safety Practices for Air Moving Devices", which is included as part of every shipment. Additional copies can be obtained by writing to New York Blower Company, 7660 Quincy St., Willowbrook, IL 60527.

ELECTRICAL DISCONNECTS

Every motor driven fan should have an independent disconnect switch to isolate the unit from the electrical supply. It should be near the fan and must be capable of being locked by maintenance personnel while servicing the unit, in accordance with OSHA procedures.

MOVING PARTS

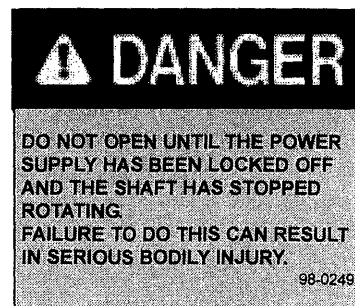
All moving parts must have guards to protect personnel. Safety requirements vary, so the number and type of guards needed to meet company, local and OSHA standards must be determined and specified by the user. Never start a fan without having all safety guards installed. Check regularly for damaged or missing guards and do not operate any fan with guards removed. Fans can also become dangerous because of potential "windmilling", even though all electrical power is disconnected. Always block the rotating assembly before working on any moving parts.

SOUND

Some fans can generate sound that could be hazardous to exposed personnel. It is the responsibility of the system designer and user to determine sound levels of the system, the degree of personnel exposure, and to comply with applicable safety requirements to protect personnel from excessive noise. Consult **nyb** for fan sound power level ratings.

AIR PRESSURE AND SUCTION

In addition to the normal dangers of rotating machinery, fans present another hazard from the suction created at the fan inlet. This suction can draw materials into the fan where they become high velocity projectiles at the outlet. It can also be extremely dangerous to persons in close proximity to the inlet, as the forces involved can overcome the strength of most individuals. Inlets and outlets that are not ducted should be screened to prevent entry and discharge of solid objects.



DANGER

DO NOT OPEN UNTIL THE POWER SUPPLY HAS BEEN LOCKED OFF AND THE SHAFT HAS STOPPED ROTATING.
FAILURE TO DO THIS CAN RESULT IN SERIOUS BODILY INJURY.

98-0249

ACCESS DOORS

The above **DANGER** decal is placed on all **nyb** cleanout doors. These doors, as well as access doors to the duct system, should never be opened while the fan is in operation. Serious injury could result from the effects of air pressure or suction.

Quick-opening doors must have the door handle bolts securely tightened to prevent accidental or unauthorized opening. Bolted doors must be tightened for the same reason.

RECEIVING AND INSPECTION

The fan and accessories should be inspected on receipt for any shipping damage. Turn the wheel by hand to see that it rotates freely and does not bind. If dampers or shutters are provided, check these accessories for free operation of all moving parts.

F.O.B. factory shipping terms require that the receiver be responsible for inspecting the equipment upon arrival. Note damage or shortages on the Bill of Lading and file any claims for damage or loss in transit. **nyb** will assist the customer as much as possible; however, claims must be originated at the point of delivery.

HANDLING AND STORAGE

Fans should be lifted by the base, mounting supports, or lifting eyes only. Never lift a fan by the wheel, shaft, motor, motor bracket, housing inlet, outlet, or any fan part not designed for lifting. A spreader should be used to avoid damage.

On direct drive Arrangement 7 or 8 fans, lifting holes are provided in the motor base to assist in handling the fan assembly. These lifting holes should be used in conjunction with the lifting eyes when lifting and positioning the fan onto its foundation. A heavy round steel bar or appropriate fixture can be passed through the lifting holes to simplify attachment of the lifting device. Be sure to follow all local safety codes when moving heavy equipment.

Whenever possible, fans and accessories should be stored in a clean, dry location to prevent rust and corrosion of steel components. If outdoor storage is necessary, protection should be provided. Cover the inlet and outlet to prevent the accumulation of dirt and moisture in the housing. Cover motors with waterproof material. Refer to the bearing section for further storage instructions.

Check shutters for free operation and lubricate moving parts prior to storage. Inspect the stored unit periodically. **Rotate the wheel by hand every two weeks to redistribute grease on internal bearing parts.**

FAN INSTALLATION

nyb wheels are dynamically balanced when fabricated. Fully assembled fans are test run at operating speeds to check the entire assembly for conformance to **nyb** vibration limits. Nevertheless, all units must be adequately supported for smooth operation. **Ductwork or stacks should be independently supported as excess weight may distort the fan housing and cause contact between moving parts.** Where vibration isolators are used, consult the **nyb** certified drawing for proper location and adjustment.

Slab-Mounted Units

A correctly designed and level concrete foundation provides the best means of installing floor-mounted fans. The mass of the base must maintain the fan/driver alignment, absorb normal vibration, and resist lateral loads. The overall dimensions of the concrete base should extend at least six inches beyond the base of the fan. The weight of the slab should be two to three times the weight of the rotating assembly, including the motor. The foundation requires firmly anchored fasteners such as the anchor bolts shown in Figure 1. Hammer-drilled expansion fasteners can be used in less demanding applications.

Move the fan to the mounting location and lower it over the anchor bolts, leveling the fan with shims around the bolts. Fasten the fan securely. When grout is used, shim the fan at least 3/4-inch from the concrete base. (See Figure 1.) When isolation is used, check the **nyb** certified drawing for installation instructions.

Elevated Units

When an elevated or suspended structural steel platform is used, it must have sufficient bracing to support the unit load and prevent side sway. The platform should be of welded construction to maintain permanent alignment of all members.

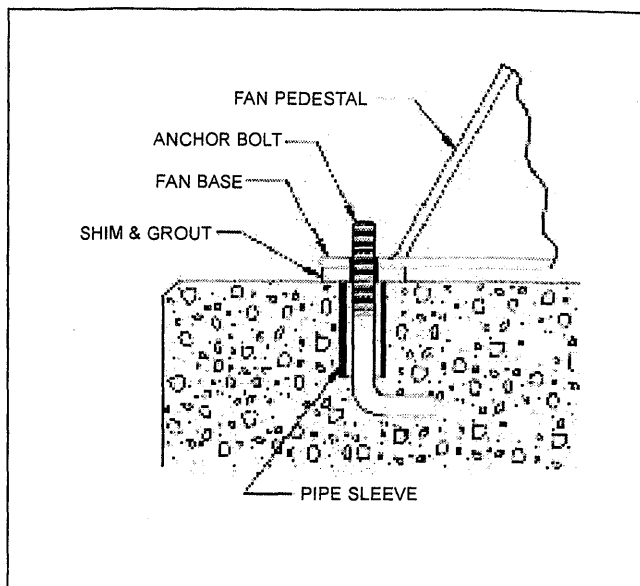


Figure 1

V-BELT DRIVE

Installation

1. Remove all foreign material from the fan and motor shafts. Coat shafts with machine oil for easier mounting. Mount the belt guard backplate at this time if partial installation is required prior to sheave mounting.
2. Mount sheaves on shafts after checking sheave bores and bushings for nicks or burrs. Avoid using force. If resistance is encountered, lightly polish the shaft with emery cloth until the sheave slides on freely. Tighten tapered bushing bolts sequentially so that equal torque is applied to each.
3. Adjust the motor on its base to a position closest to the fan shaft. Install belts by working each one over the sheave grooves until all are in position. Never pry the belts into place. On **nyb** packaged fans, sufficient motor adjustment is provided for easy installation of the proper size belts.
4. Adjust sheaves and the motor shaft angle so that the sheave faces are in the same plane. Check this by placing a straightedge across the face of the sheave. Any gap between the edge and sheave faces indicates misalignment. Important: This method is only valid when the width of the surface between the belt edge and the sheave face is the same for both sheaves. When they are not equal, or when using adjustable-pitch sheaves, adjust so that all belts have approximately equal tension. Both shafts should be at the right angles to the center belt.

Belt Tensioning

1. Check belt tension with a tensioning gage and adjust using the motor slide base. Excess tension shortens bearing life while insufficient tension shortens belt life, can reduce fan performance and may cause vibration. The lowest allowable tension is that which prevents slippage under full load. Belts may slip during start-up, but slipping should stop as soon as the fan reaches full speed. For more precise tensioning methods, consult the drive manufacturer's literature.
2. Recheck setscrews, rotate the drive by hand and check for rubbing, then complete the installation of the belt guard.